Competitor research

Prodigy Math

Overview:

A fantasy RPG where students create wizard avatars, explore magical worlds, and engage in turn-based monster battles by answering math questions.

Target Audience: Grades 1–8 (Ages 6–14)

What they do well:

- Highly engaging visual world-building and avatar customization.
- High student engagement and motivation to log in daily.

Limitations:

- Uses extrinsic gameplay integration ("chocolate-covered broccoli" - a design metaphor in educational gaming that describes wrapping boring, repetitive educational tasks (the "broccoli") inside an unrelated, fun video game wrapper (the "chocolate")).
- Fails to cover high school learners (Grades 9–11); the visual style feels too childish for teenagers, and its simple turn-based system cannot support complex upper-grade algebra.
- Abruptly pauses gameplay with pop-up quiz cards, turning math into a progress tollgate rather than a core game mechanic.

Competitor research

IXL Math

Overview:

A skill-focused learning platform offering structured math drills categorised by grade level and educational standards.

Target Audience: Pre-K to Grade 12 (Ages 4–18)

What they do well:

- Comprehensive curriculum coverage with granular skill tracking.
- Clear, actionable progress metrics for educators and parents.

Limitations:

- Uses punitive scoring mechanics (slashing progress scores on wrong answers), increasing math anxiety.
- Displays static post-mortem solution text only after a student misses a problem, failing to guide them through steps mid-problem.
- Lacks any RPG elements or story wrapper, feeling like a high-stress digital worksheet for both primary (Grades 1–6) and secondary (Grades 7–11) students.

Competitor research

Monster Math

Overview:

An action-adventure arcade game where players solve arithmetic problems to defeat monsters and progress through levels.

Target Audience: Grades K–3 / K–5 (Ages 5–10)

What they do well:

- Integrates basic mental arithmetic directly into active arcade gameplay.
- Helps young children build speed in basic math facts.

Limitations:

- Relies on hardcoded, non-adaptive question pools that lead to quick answer memorization and high user churn.
- Encourages answer memorization over deep conceptual understanding, causing content to run dry quickly.
- Locked strictly to early primary arithmetic; completely ignores Intermediate, Senior, and High School mathematics (Grades 4–11 algebra, geometry, and functions).

Secondary Research

Global Context: Math Anxiety & Cognitive Performance

High Prevalence of Math Anxiety:

Globally, over **39%** of **15-year-old students** report feeling helpless, severe apprehension, or tension when dealing with mathematics problems (OECD, 2023).

Working Memory Degradation:

Cognitive research demonstrates that math anxiety acts as an internal distraction that temporarily degrades a student's working memory capacity during problem-solving (Ashcraft, 2002).

The Avoidance Cycle:

When working memory is compromised by anxiety, standard arithmetic feels significantly harder than it actually is, creating a self-fulfilling cycle of academic failure and practice avoidance (Maloney & Beilock, 2012).

Secondary Research

Local Context: South African CAPS Curriculum Participation & Performance

Low Pure Mathematics Participation:

Official National Senior Certificate (NSC) data shows that only 34.1% of South African matric candidates write pure Mathematics, with the majority opting for Mathematical Literacy (Department of Basic Education [DBE], 2026).

Gateway Subject Performance Drop:

Among learners who write pure Mathematics, the national pass rate declined from 69% in 2024 to 64% in 2025, alongside a ~50% drop in distinction rate (DBE, 2026).

Senior Phase Bottlenecks (Grades 10–12):

Educational researchers highlight that dropout and subject-switch pressure intensifies sharply between Grades 10 and 12, driven by unresolved foundational math gaps from earlier primary/ intermediate grades (Naicker, 2026).

Secondary Research

Designing for Dyscalculia

Clear, Large, Well-Spaced Numbers:

No special font is proven to help. Larger digits improved calculation fluency in a study of 436 students (Takahashi & Inoue, 2022), and leaving space around numbers reduces visual distortion (Parker, 2022).

Visual Number Models:

Targeted number-line training improved number representation and calculation skills in learners with maths learning disabilities (Marcon & Lafay, 2019).

Short, Adaptive Game Sessions:

Calcularis, an adaptive maths game played for 20 minutes a day, significantly improved number representation and arithmetic in children with dyscalculia (Käser et al., 2013).

Secondary Research

Gamification in Learning: Does It Help?

Better Learning Results:

A meta-analysis of gamified learning studies found a medium positive effect on what learners actually learned (g = 0.49), not only on how much they enjoyed it (Sailer & Homner, 2020).

Higher Motivation & Engagement:

The same analysis found gamification also raised learners' motivation (g = 0.36) and how actively they took part in learning tasks (g = 0.25) (Sailer & Homner, 2020).

Story & Play Matter Most:

Game fiction (a story world to play in) and social play were the game elements that most improved how learners engaged with tasks (Sailer & Homner, 2020).

Secondary Research

A Game You Learn In, Not a Gamified Learning App

Games Beat Non-Game Lessons:

A review of digital learning games for K–16 learners found games improved learning compared with non-game teaching (g = 0.33), and that game design choices changed how well they worked (Clark et al., 2016).

Maths as the Game Mechanic:

In Zombie Division, a maths game for 7–11 year olds, children learned more when the maths was built into the core gameplay than when the game was just a reward between maths questions (Habgood & Ainsworth, 2011).

7× Longer Play Time:

When children could choose freely, they played the version where the maths was the gameplay about 7 times longer than the version that used the game as a reward (Habgood & Ainsworth, 2011).

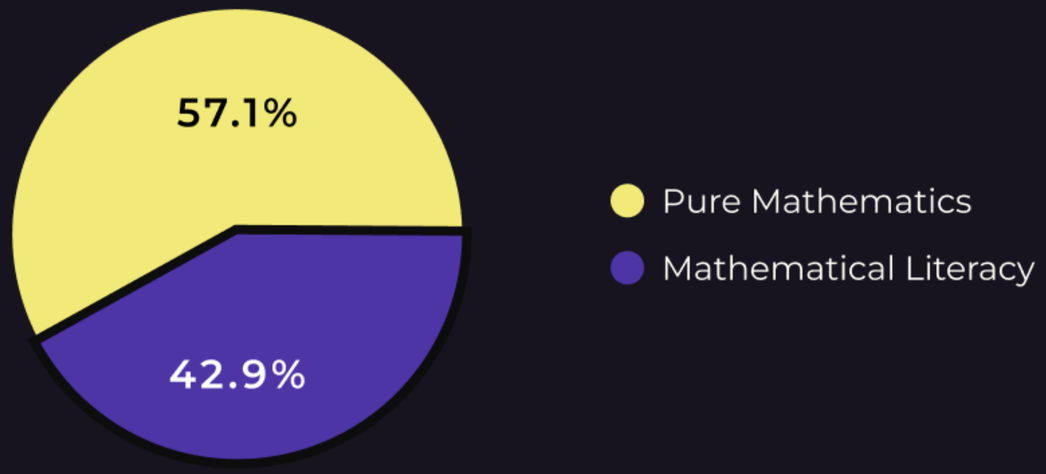
Why Tower of Arithmetic is a game first:

Most maths apps add points and badges on top of a quiz. My target users are gamers, so I made the maths the game itself: every attack, spell and puzzle is solved with maths. The research shows this keeps learners playing longer and learning more.

Survey research

Question 1:

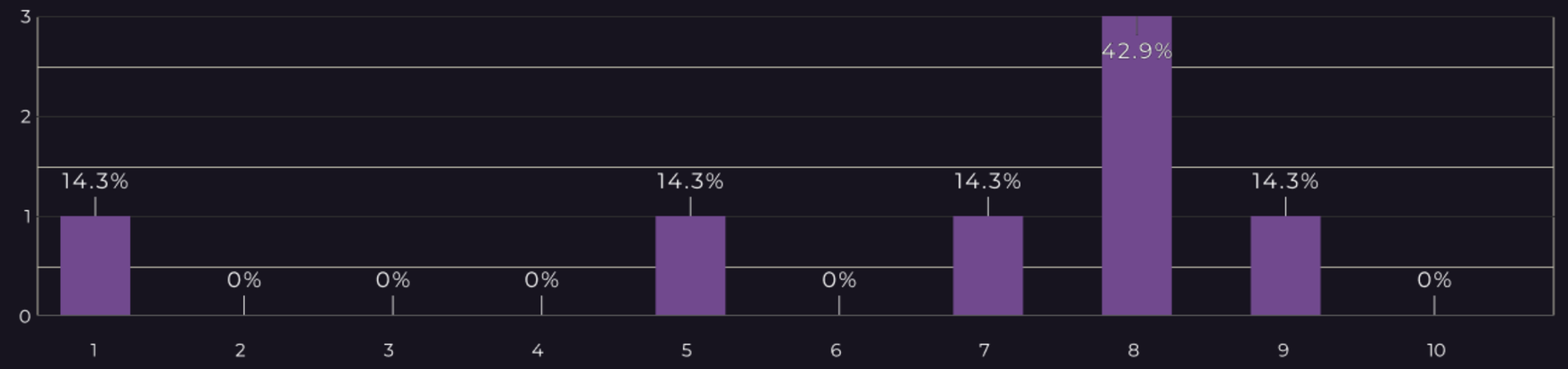
Which math subject of math did you take?



Survey research

Question 2:

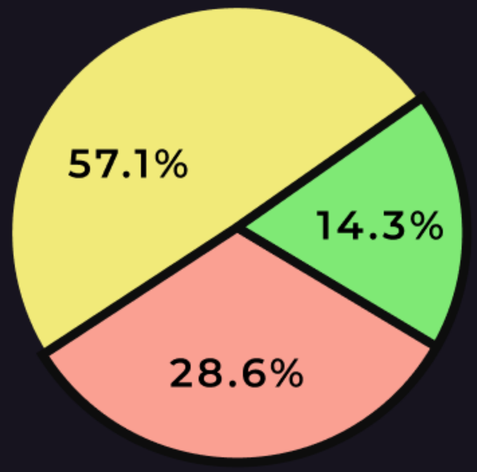
On a scale of 1 to 10, how much do (or did) you struggle with Pure Mathematics in school?



Survey research

Question 3:

When working through a math problem, where do you get stuck the most in the process?

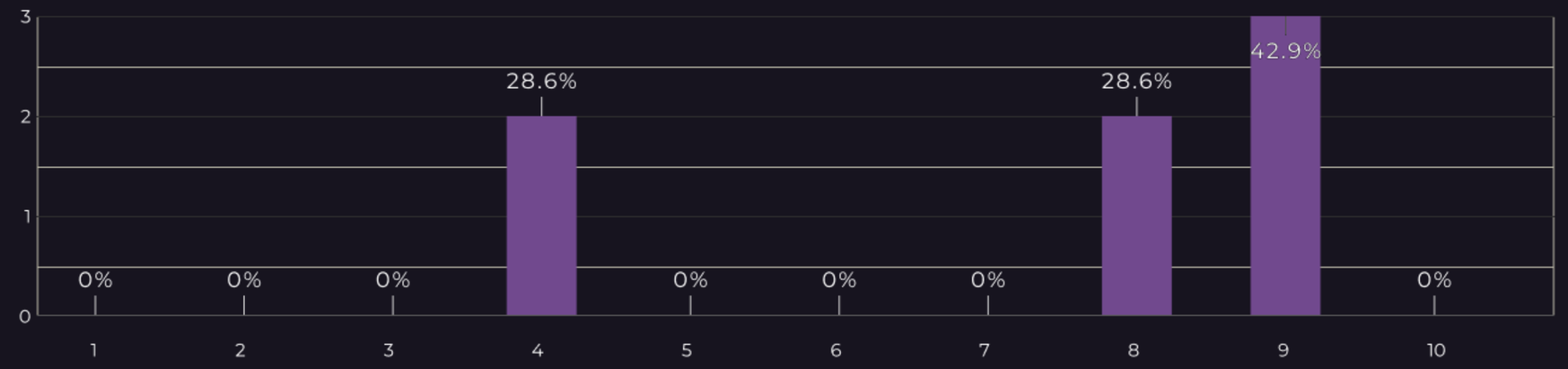


- Understanding the Core Concept: I don't understand why the concept or formula works.
- Applying Theory to Action: I know the formula/rules, but I don't know how to start or implement the steps on paper.
- Remembering Rules: I forget the exact steps or formulas needed for the problem.
- Some of the questions were bullshit. And other spots were minor mistakes. Geometry

Survey research

Question 4:

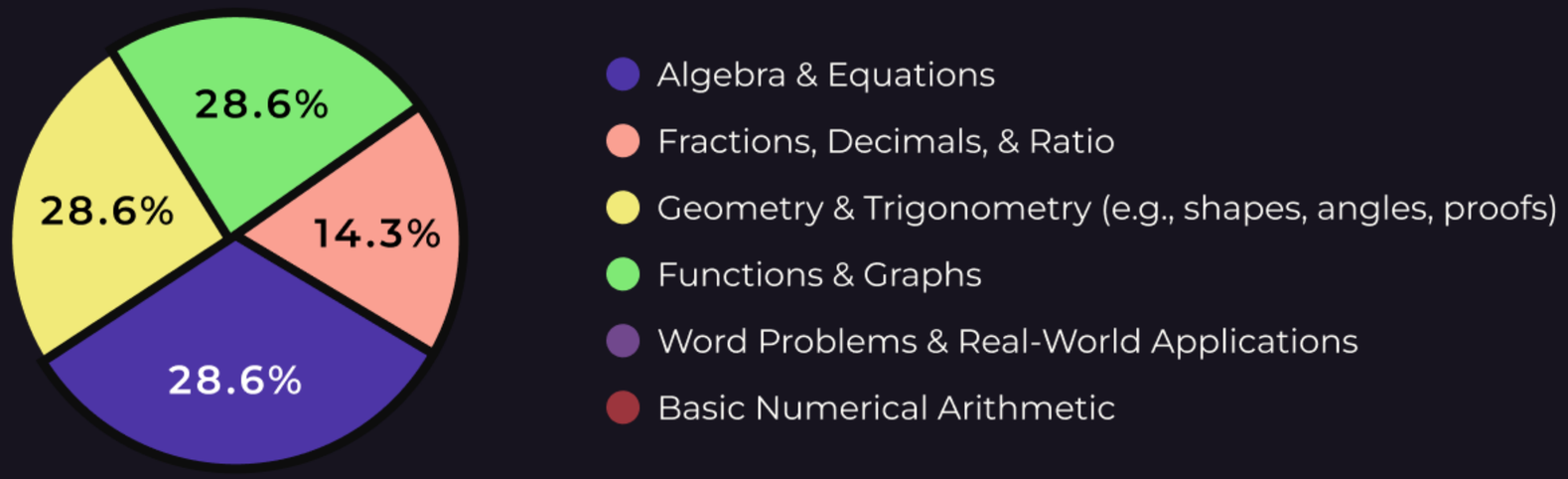
On a scale of 1 to 10, how much Math Anxiety (stress, tension, or freezing up) do you experience during math tests or practice?



Survey research

Question 5:

Which specific math topic area causes you the biggest mental hurdle or frustration?



Survey research

Question 6:

Would you prefer to practice math through an interactive game rather than traditional worksheets, and why?

Yes! I think that gamification for math could be a great way to learn, especially when it shows you exactly where you went wrong

Yes. I think making it a game will help with remembering the steps in a more fun way

Yes to help remember application

I would prefer an interactive game but only if it was visually enticing - I'm a visual learner so I struggle with math because there isn't much visual to it at all

Like for example - For business subjects I could draw people and situations to learn the different ways of handeling conflict in the company

but math were numbers sooo I struggled with that so maybe something visual with this digital solution would really help, especially creative people !

YES please because I think it can create a fun, memorable experience that can also benefit in remembering the mathematical rules as well.

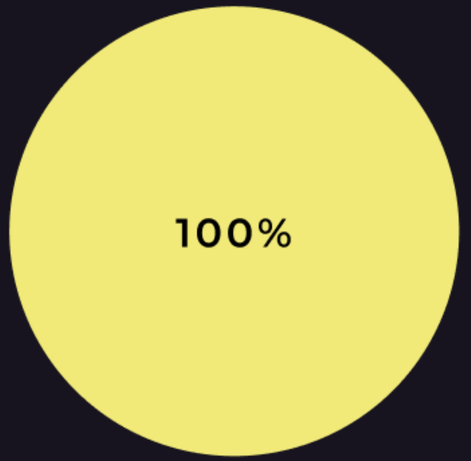
Yes, I think it would make learning more fun and help me overthink less.

Yes. More fun more learning

Survey research

Question 7:

In your opinion, how should a gamified math app fit into your learning?

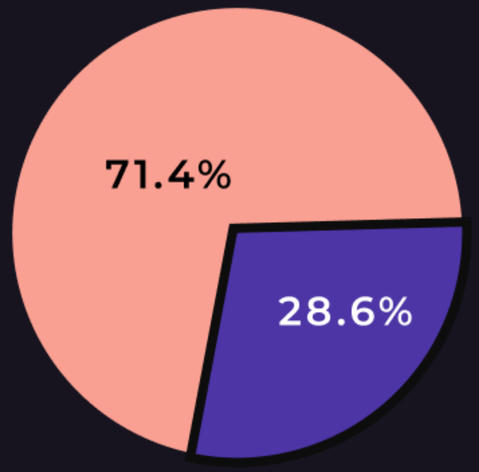


- Practice & Reinforcement: It should focus on practicing topics I have already been taught in class.
- Learning from Scratch: It should try to teach me brand-new math topics completely from scratch before school does.
- Both: It should allow me to learn new topics and practice old ones.

Survey research

Question 8:

If a math game presents a topic you haven't learned in school yet, what is the best way to introduce it without making you feel overwhelmed?

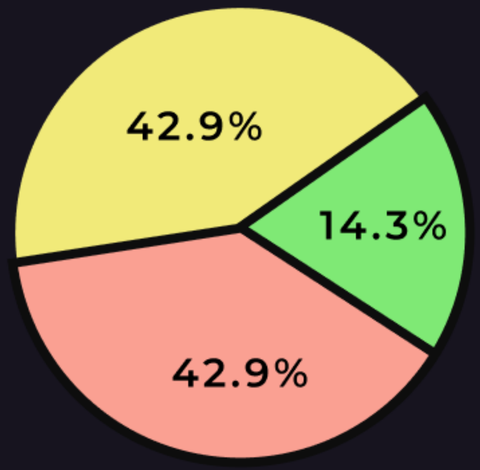


- A step-by-step worked example showing how to solve one sum.
- A short 30-second interactive visual guide before the question.
- A "Need Help?" button that gives a clue only if I ask for it.
- Do not show me topics I haven't learned in class yet.

Survey research

Question 9:

When you make a mistake on a math problem, what kind of feedback helps you keep trying instead of giving up?

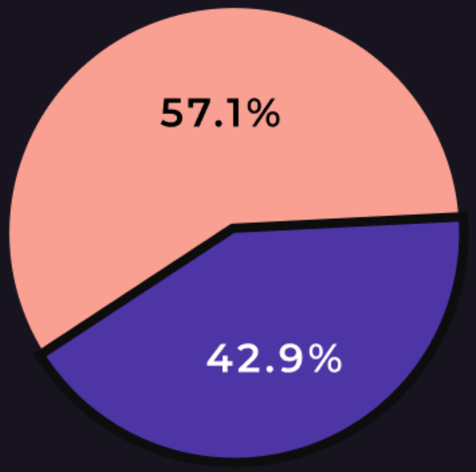


- A short, 1-sentence clue guiding my next step while I am still solving it.
- Showing the full, completed answer after I fail the whole problem. with step by step explication
- Just telling me "Incorrect" so I can figure it out on my own

Survey research

Question 10:

What is the primary reason you put off or avoid doing regular math practice/homework at home?

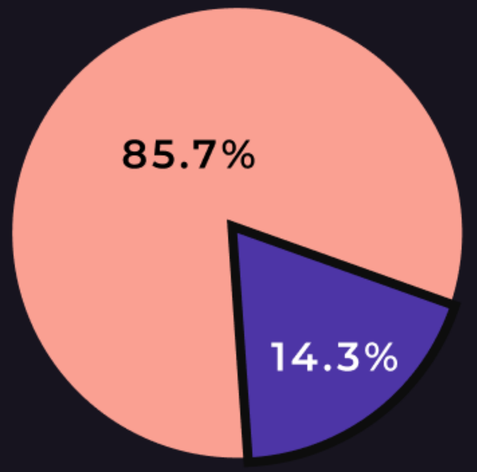


- It feels like a boring, repetitive chore (e.g., textbook worksheets).
- I get frustrated and stressed when I get stuck with no immediate help.
- It takes away too much time from gaming, hobbies, or relaxing.
- I don't see the reward or motivation behind doing it.

Survey research

Question 11:

Should time pressure (like countdown timers or fast-paced actions) be included in a math practice game?



- Yes: Timers make the game exciting and build fast mental recall
- No: Timers trigger panic and cause math anxiety.
- Optional: Timers should only exist as a separate "Speed Mode" that can be turned off.

Survey research

Question 12:

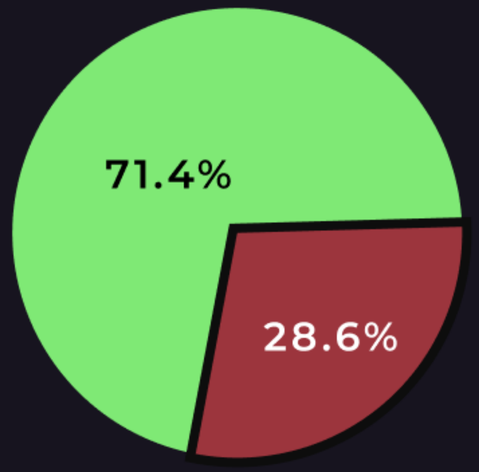
How do countdown timers or timed math tasks affect your performance?



Survey research

Question 13:

If you make an error or get stuck during a battle, which penalty/feedback system would keep you motivated instead of feeling frustrated?

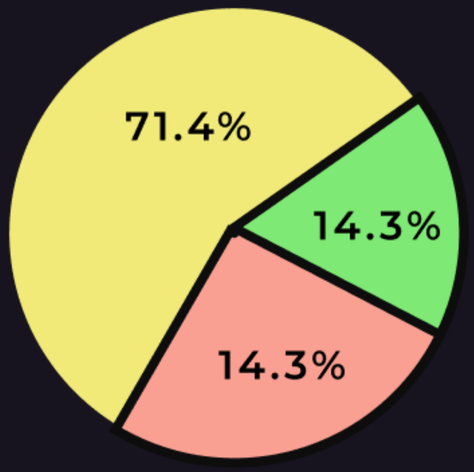


- Your attack fails, you get a short explanation of what you did wrong, and the opponent's turn begins (taking damage before your next attempt).
- You receive an "Incorrect" notice with a Socratic clue explaining your exact mistake, but you remain on your turn to try solving the same sum again.

Survey research

Question 14:

How would you prefer game mechanics to connect to solving math problems?

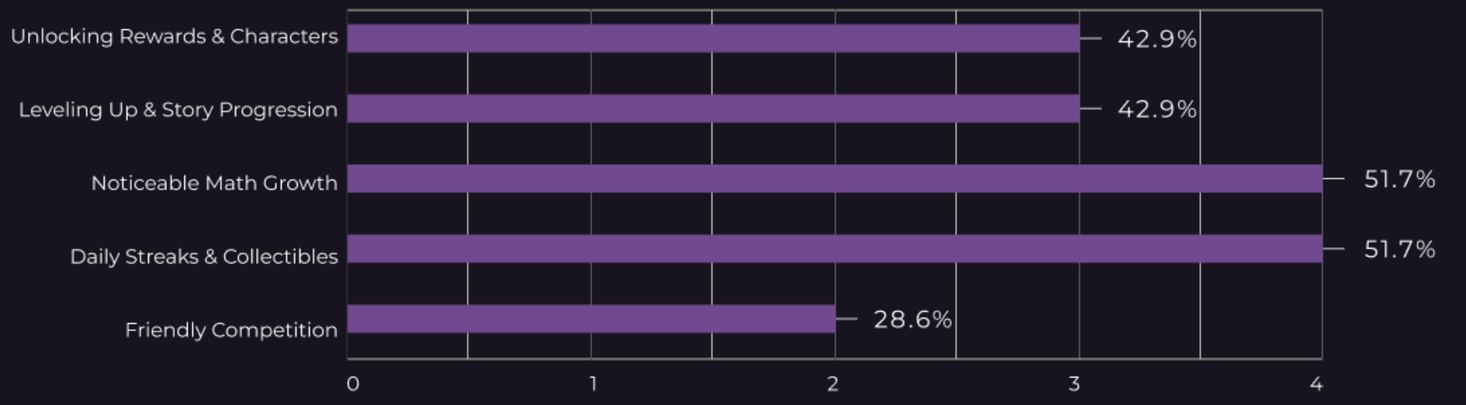


- Solving an equation directly casts a spell or disarms a trap (Math = The Action).
- Reward System: Solving sums gives me gold/XP to buy armor, skins, and character upgrades after the battle.
- Progression: Solving sums lets me unlock new floors, bosses, and story chapters.
- All of the above

Survey research

Question 15:

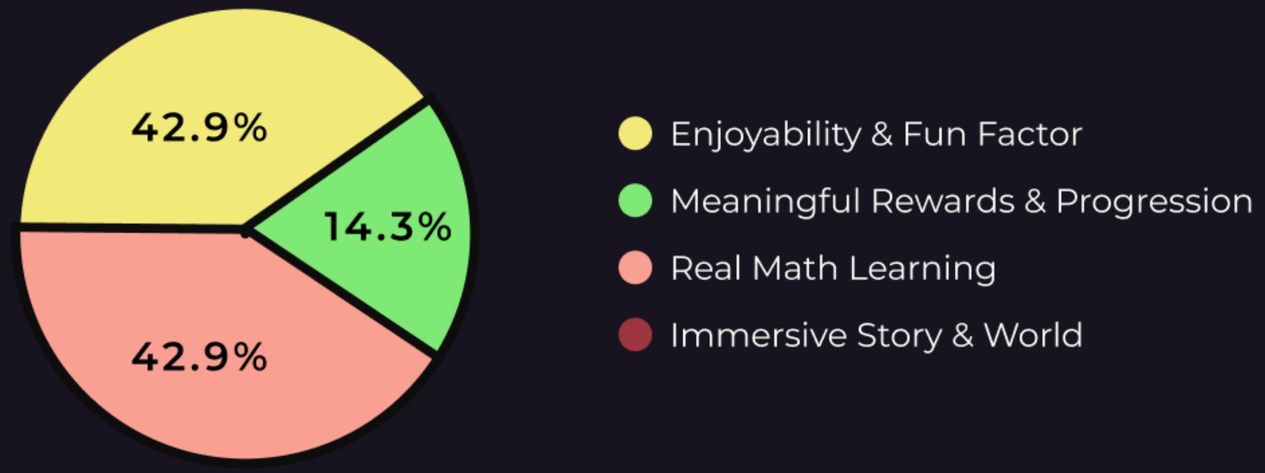
What would be the BIGGEST motivation for you to keep coming back to play a math RPG regularly on your own?



Survey research

Question 16:

If you could only choose ONE, what is the single most important thing in a gamified math app to make it worth your time?



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